



ENERGETIKA
ENERGETICS

Published from 2004
Ministry of Press and Information
of Azerbaijan Republic,
Registration number 3337, 07.03.2011

ISSN 1816-2126
Number 01, 2025
Section: English

ECOENERGETICS

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Water Degradation Challenges in Azerbaijan's Karabakh Region

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Abstract. The Karabakh region, located in the eastern and southeastern ranges of the Lesser Caucasus and belonging to Azerbaijan, has great historical and geographic significance. However, following the dissolution of the Soviet Union, Karabakh declared independence from Azerbaijan in a move that was not recognized by Baku. This was compounded by decades of illegal military occupation by Armenia, in breach of international law and Azerbaijan's inviolable borders. The lengthy occupation caused severe environmental and economic damage, ravaging for the region ecosystems and undermining the economic sustainability of the territory. The thesis provides a comprehensive assessment of the environmental devastation caused in Karabakh and surrounding regions over almost thirty years under occupation. The study examines a wide swath of environmental factors, especially in water resources. The main task of the study is to reveal the ecological imbalance caused by the war and to highlight the need for restoring the natural balance and the sustainable coexistence of society and nature. Political-military science or human-environmental relations in the context of restoration of devastated ecosystems and restoration of human-environmental complexes destroyed under the influence of long-term military actions are also surveyed in the research. The study describes a broad spectrum of environmental harms, from waterway pollution to biodiversity loss, resource extraction and soil erosion. They advise in this piece that restoration efforts — methodical, collective — be undertaken to restore the ravaged landscapes. The thesis presents a comprehensive overview of the environmental cost of the Karabakh territory through the "integrated" use of ecological and geographic assessment, statistics, field observations and satellite images.

Keywords: Azerbaijan, Nagorno-Karabakh, liberated regions, ecological challenges, resource potential, biological diversity

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1. Introduction

Karabakh (Azerbaijani: Qarabağ) lies in the mountain and foothill part of the eastern and southeastern parts of the Azerbaijani side of the Lesser Caucasus, sloping to the Karabakh Plain. It is one of the historical-geographical and ancient regions of Azerbaijan. Following the break-up of the USSR, the region illegally proclaimed independence from Azerbaijan, violating international law and the officially determined borders of the state.

Azerbaijan's fertile natural setting, and its socio-economic status, provide the basis for a wide range of national economic activities. These include heavy industry and other services sectors such as trade, health, public eating, recreation, education, and culture—all financed by the country's rich natural resources [2].

This research paper aims to analyze the state of economic development of the given region's infrastructure in the years where it was unlawfully occupied by Armenian forces, paying special attention to the period of utter neglect. This depravity was systematically maintained until these territories were liberated in 2020. Due to the occupation of Karabakh and seven neighboring districts, Azerbaijan was unable to fulfill its international commitments in environmental protection and the rational management of resources for nearly three decades.

The aim of the paper is to examine the tremendous environmental degradation the Karabakh region underwent with the occupation and to introduce today's rehabilitation process focusing on recovery of ecosystems as well as sustainability of natural use in the area liberated. By blending theoretical paradigms and pragmatic research approaches, the study seeks to offer a critical evaluation of the issues as well as the envisaged solutions concerning the rebuilding of ecological balance and harmonious coexistence between nature and society after decades of war [1-5].

2. Experimental detail

The research draws on diverse sources including the institute archives of the Institute of Geography of the Ministry of Science and Education of Azerbaijan (Imrani et al., 2021; 2022; Eminov et al., 2024), outcomes of ecological activities of the Faculty of Ecology and Soil Science at Baku State University (Ibrahimov & Sadullayev, 2023), and research conducted by the Faculty of Geography of the same university (Dargahov et al., 2023). Other sources cited are M. Museyibov's data on landscapes (2014), research on environmental awareness and eco-consumerism in

Ukraine (Vysotska et al., 2021; Vysotska & Vysotskyi, 2022), and the author's own data for the 2003-2023 period (Bayramova, 2018, 2019, 2023).

The study also utilizes National Aerospace Agency of Azerbaijan data (Suleymanov, 2015) and the State Committee "AzerCosmos" (2019–2022) data and respective national and international websites and databases data.

Methodologically, the study is carried out according to an ecological-geographical analytical plan involving the exploration of available archives and literature, employing theoretical and methodological tools, applying geographical determinism (Getman et al., 2023; Vysotskyi et al., 2022; Vysotskyi et al., 2023), making statistical summations, conducting field investigation and expeditions, utilizing cartographic instruments, and applying monitoring and forecasting strategies. The author further applies a subjective perspective when analyzing the geo-ecological potentiality of natural resources within the newly occupied territory of Azerbaijan. [2]

This research deals with the environmental and legal situation of environmental issues on Azerbaijan's liberated territories, specifically addressing the recovery of Karabakh's natural specifics. The natural resources of these lands were extensively over-exploited and devastated from 1988 to 2020 by inhabitants of a neighboring state and their external allies. This research presents validated satellite photos establishing the extent of landscape degradation and advocating the restoration of ecological balance in the study areas. The impacts of the illegal occupation and illegal natural resource exploitation go beyond economic harm—they have unleashed catastrophic environmental repercussions, including:

- Water bodies pollution (and, in most cases, poisoning),
- Loss of biodiversity,
- Illicit mining of mineral reserves,
- Soil erosion and loss,

These impacts have significantly destabilized the environmental equilibrium of the region.

These impacts have significantly destabilized the ecological equilibrium of the region. The following goals are set by this study:

- Ecological and geographical assessment of the natural resources potential of the occupied territories of Azerbaijan;
- Restriction of natural and territorial structuring of the study area by geographic and ecological factors;
- Evaluation of the extent of environmental destruction of natural habitats by military occupation
- Utilize reconstruction work for the infrastructure to enable ecological rehabilitation, ecologically friendly and sustainable natural resource management in the future after the occupation.

2.1 Harm to Azerbaijan's Unique Natural Landscape

The degradation of these ecologically rich areas risks the entire South Caucasus region more generally. Main points incorporated in below:

- Significant environmental policy analyses of natural resource management and conservation of biodiversity identify that such destructive conduct is a blatant violation of international law.
- Legally, no lawful transfer of control of the occupied territories was done. This is as confirmed by the 2016 report of the Ministry of Foreign Affairs of Azerbaijan, titled *Illegal Economic and Other Activities in the Occupied Territories of Azerbaijan: Evidence from Satellites*. Military occupation has been acknowledged by international law as being temporary and illegal to diplomatically recognize sovereignty over the occupied territory by the occupying state.
- Satellite photographs released by the State Committee AzerCosmos in its 2019 report further indicate the scope of damage caused to Azerbaijan's ancient lands due to illegal economic practice and ecological degradation [4].

2.2 Water Resources Damages to Karabakh's Natural Ecosystems:

During the occupation by Armenians, Karabakh water sources were diverted and polluted. Azerbaijan relies on rivers that have their sources in Türkiye, Armenia, and their traditional homelands, such as Karabakh.

The highland lakes of the Lesser Caucasus, such as Boyuk Alagel and Kichik Alagel, also were subjected to the same deterioration. A lot of water infrastructure was destroyed or damaged over the years, including:

- A 6,426 km irrigation network
- There are eight reservoirs with a total capacity of 640 million m³,
- About 200 km of drainage and collector networks,
- More than 1,400 underground water sources,
- 88 pumping stations and over 500 hydraulic engineering structures,
- 220 small-scale regional hydropower plants.

All of the necessary systems had been for a long time neglected or willfully destroyed in the process of occupation [5].

There were about a dozen reservoirs, including the large Sarsang Reservoir on the Terter River, that were held by Armenian military. With a capacity of some 500 million cubic meters, Sarsang was initially constructed as a massive hydraulic facility with the ability to irrigate over 100,000 hectares of land on the Karabakh plain. However, following decades of neglect and no maintenance of the facilities under occupation, nearly 400,000 individuals who lived in the lower foothills and low-lying areas experienced grave water shortages. The Sarsang Reservoir was at the middle of humanitarian, environmental, and man-made emergencies since Azerbaijan regularly reported to international bodies regarding the threats. Armenian forces used the reservoir's water resources as a political tool of abuse: During autumn and winter, when water use was low, the annual release of 85–90% of the reservoir came suddenly, flooding Azerbaijani villages downstream. In Summer peak agricultural demand, only 10–15% of the total annual quota of water was let out, leading to severe irrigation water shortages with ruinous effects on agricultural production downstream [1].



Figure 1. Satellite imagery by Azersky. Sugovushan water reservoir

This intentional manipulation of water flow created **widespread environmental stress** and pushed local farming to a standstill. Notably, the construction of the Sarsang Reservoir was originally funded by Azerbaijan's national budget in the 1970s, representing a major investment in regional development [5].

The resolution condemned Armenia's actions as a deliberate use of water as a means of political pressure and emphasized the urgency to:

- Allows independent hydrologists to access the area,
- Ending the misuse of water resources,
- Appreciating the dangerous condition of the dam, which could result in a huge human tragedy if left unaddressed.

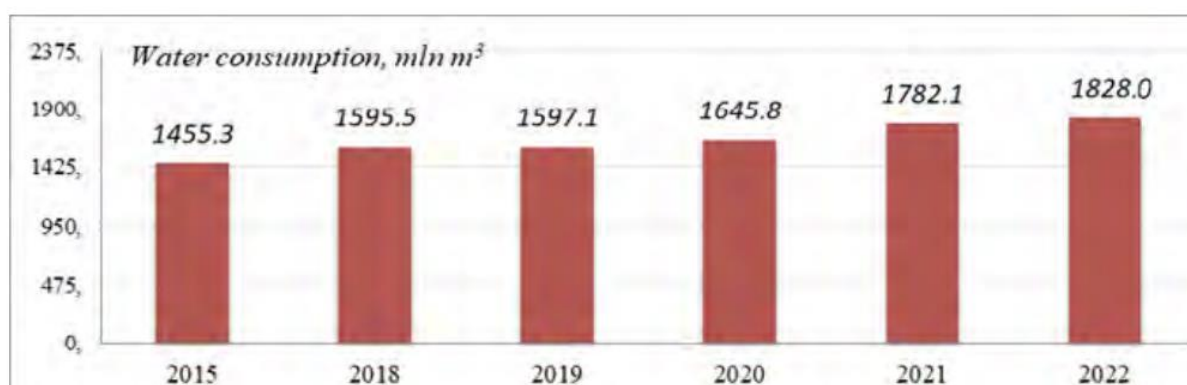


Figure. 2. Consumption of water in de-occupied areas of Azerbaijan

3. Conclusion

Occupation of Karabakh not only caused political and humanitarian issues, but also environmental degradation in usage and destruction of water resources. Pollution, diversion, and neglect of rivers, lakes, and reservoirs deeply destabilized the balance of nature of the area and its agrarian potential. The Sarsang Reservoir, once a symbol of regional development, was transformed into a tool of environmental and political pressure, harming nature and human

Y.R.Yusifova, Ecoenergetics, vol. 30, № 1, pp.100-103, (2025)

existence. This study highlights the imperatives of the strategic rejuvenation of the water infrastructure and ecosystem balance of Karabakh. Sustainable renaissance would require vigorous political willpower, environmental planning, modern technologies, and regional collaborative endeavors. Only through these collective efforts would the water systems of Karabakh rejuvenate and be safeguarded for generations to come.

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A photograph of a wind farm with several white wind turbines on a green, hilly landscape under a cloudy sky. The image is framed by a dark green curved border at the top.

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A graphic of a water splash or wave, rendered in a light green and white color scheme, positioned on the right side of the lower half of the cover.